SEng6241:- Mobile Application Development

Chapter 1: Introduction

"I'm going to destroy Android, because it's a stolen product. I'm willing to go thermonuclear war on this."
—Steve Jobs, Apple Inc.

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Contents

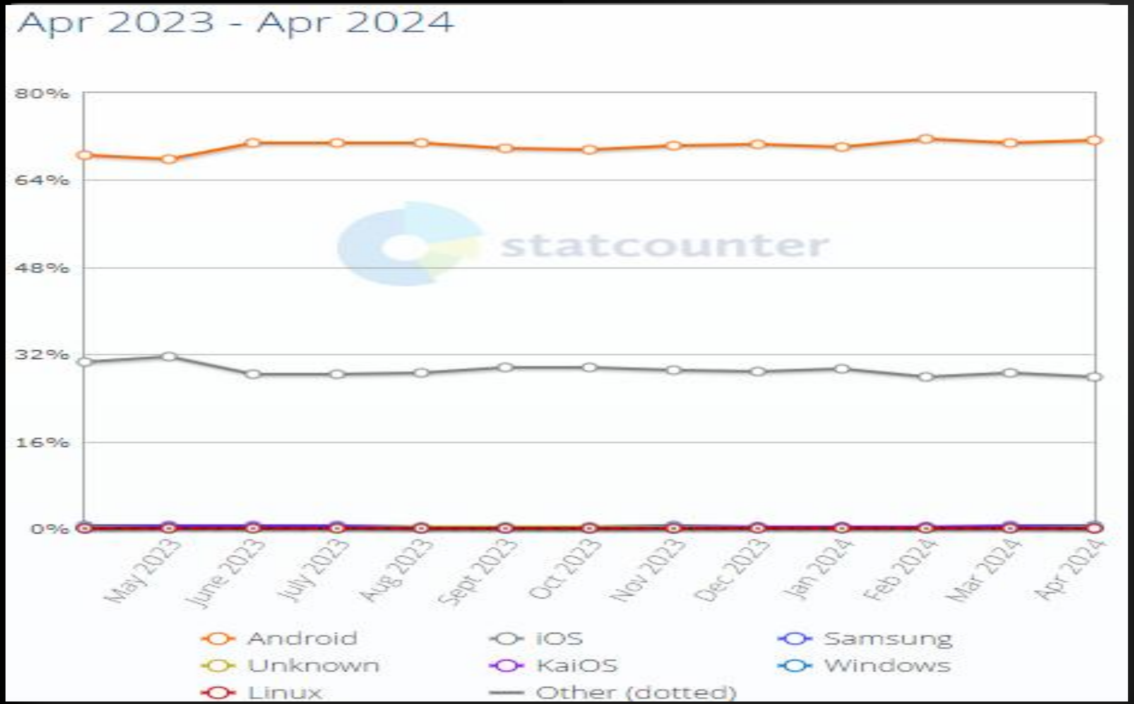
- Introduction to Mobile Computing
  - Android Development Environment

Mobile Computing

- *Mobile computing has transformed society, enabling seamless communication, access to information, and innovative applications.*
- *It has transformed the global landscape, revolutionizing the mobile market. From smart phones to tablets, it has reshaped how we work, communicate, and access information on the go.*
- *Before Android emerged as a dominant force in the mobile industry, several challenges plagued the market, hindering its growth and innovation such as*
  - *Fragmentation: A Pervasive Barrier to Unity*
  - *Closed Ecosystems: Restricting Creativity and Diversity*
  - *Limited Developer Support: Complexities in Application Development*
  - *High Development Costs: A Barrier for Innovation*
  - *Lack of Interoperability: Incompatibility Woes*
  - *Security Concerns: Vulnerabilities in a Disjointed Environment*

Mobile Operating System Market Share Worldwide

OS Type	Market Share
Android	74.3%
iOS	24.76%
KaiOS	0.21%
Samsung	0.2%
Unknown	0.19%
Windows	0.12%



January 2020

Android in Mobile Computing

- What is android?
- Android version
- Features of android
- Architecture of android
- Anatomy of an Android Application

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What is an Android ?

What is android ?

- ***Android*** *is a software package and Linux based operating system for mobile devices such as tablet computers and smart phones.*
- *It is a **software stack** for **mobile devices** that includes an operating system, middleware and key applications.*
- *Initially, **Andy Rubin** founded Android Incorporation in Palo Alto, California, United States in October, 2003.*
- *It is purchased by Google in 2005 and later the OHA (Open Handset Alliance).*
- *In 2007, Google announces the development of android OS.*

What is Open Handset Alliance (OHA) ?

Mobile companies

SAMSUNG	XIAOMI	ASUS	INFINIX
APPLE	GOOGLE	ALCATEL	ULEFONE
HUAWEI	HONOR	ZTE	TECNO
NOKIA	OPPO	MICROSOFT	DOOGEE
SONY	REALME	VODAFONE	BLACKVIEW
LG	ONEPLUS	ENERGIZER	CUBOT
HTC	VIVO	CAT	ITEL
MOTOROLA	MEIZU	SHARP	TCL
LENOVO	BLACKBERRY	MICROMAX	PANASONIC

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What is Open Handset Alliance (OHA) ?

- *It's a consortium of 84 companies such as Google, samsung, AKM, synaptics, KDDI, Garmin, Teleca, Ebay, Intel etc.*
- *It was established on 5th November, 2007, led by Google.*
- *It is committed to advance open standards, provide services and deploy handsets using the Android Plateform.*

- *The project responsible for developing the Android system is called the Android Open Source Project (AOSP) and is primarily lead by Google.*
- *In 2008, HTC launched the first android mobile.*

– *The goal of android project is to create a successful real-world product that improves the mobile experience for end users.*

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What is Android Version?

Android version

Version (Sdk Version)	Code name	API Level
1.5	Cupcake	3
1.6	Donut	4
2.1	Eclair	7
2.2	Froyo	8
2.3	Gingerbread	9 and 10
3.1 and 3.3	Honeycomb	12 and 13
4.0	Ice Cream Sandwitch	15
4.1, 4.2 and 4.3	Jelly Bean	16, 17 and 18
4.4	KitKat	19
5.0	Lollipop	21
6.0-6.0.1	Marshmallow	23
7.0-7.1.2	Nougat	24-25
8.0-8.14	Oreo	26-27
9.0	Pie	28
10-14	android 10-14	29-34

- *Android has gone through quite a number of updates since its first release and codenames and API Level provided by Google.* ***Aestro & Blender***

Open the link → ***Read more.....***

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Features of Android

Features of android

- ***Storage***: Uses SQLite, a lightweight relational database, for data storage.
- ***Connectivity***: Supports GSM/EDGE, IDEN, CDMA, EV-DO,UMTS, Bluetooth (includes A2DP and AVRCP), WiFi, LTE, WiMAX. 3G and 4G
- ***Messaging*** :Supports both SMS and MMS.
- ***Tethering***: Supports sharing of Internet connections as a wired/wireless hotspot
- ***Media support***: includes support for the following media: H.263,H.264 (in 3GP or MP4 container), MPEG-4 SP, AMR, AMR-WB (in 3GP container), AAC, HE-AAC (in MP4 or 3GP container), MP3, MIDI, OggVorbis, WAV, JPEG, PNG, GIF, and BMP

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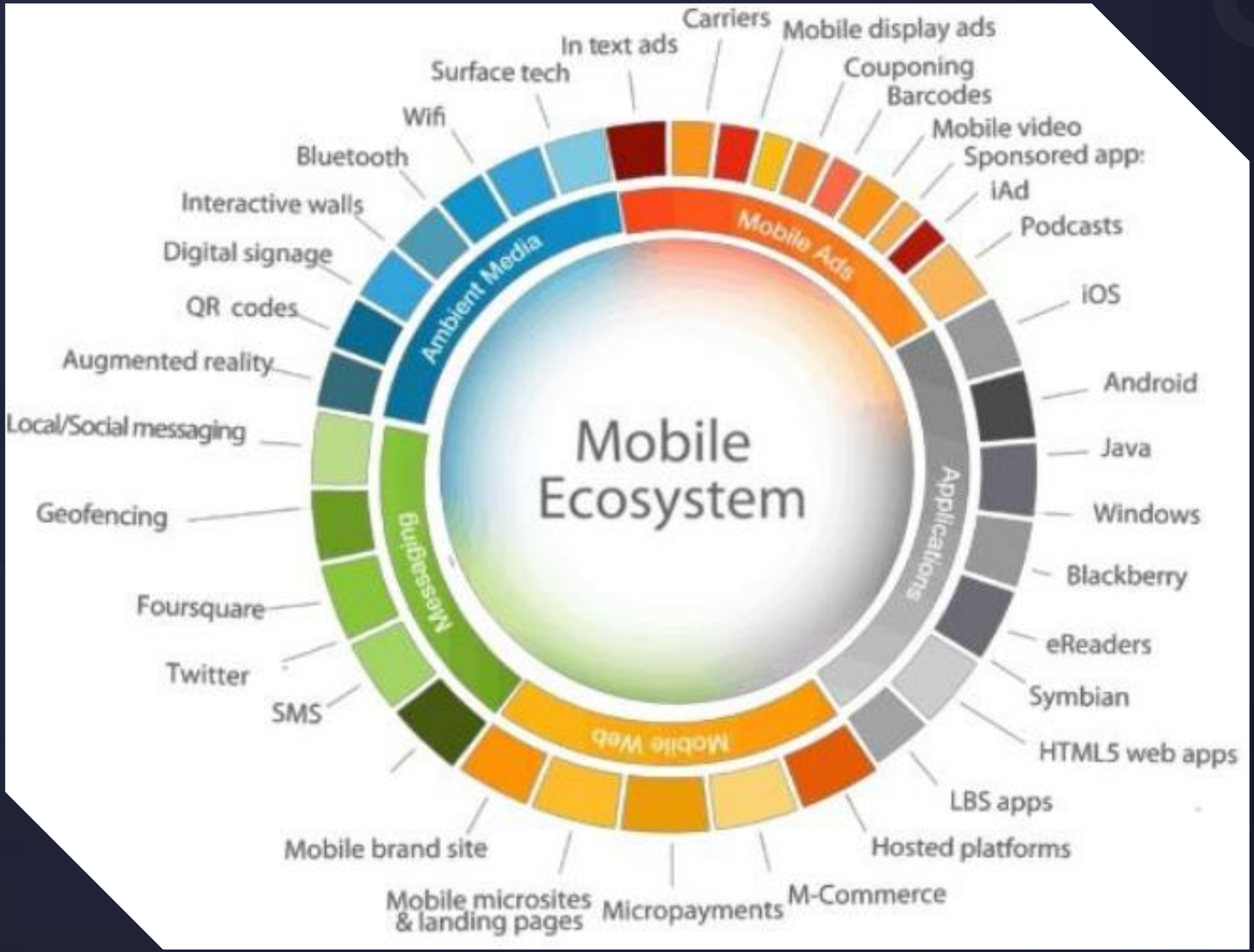
- ***GCM Google Cloud Messaging (GCM)*** *is a service that let developers send short message data to their users on Android devices, without*
  needing a proprietary sync solution.
- ***Hardware support:*** *Accelerometer Sensor, Camera, Digital Compass, Proximity Sensor, and GPS*
- ***Multi-touch*** *:Supports multi-touch screens*
- ***Multi-tasking****: Supports multi-tasking applications.*
- ***Web browser****: Based on the open-source WebKit, together with Chrome's V8 JavaScript engine.*
- *…………………..etc*

Shared features with other mobile operating system in ecosystem



Architecture of android

Architecture of android

The Android OS is roughly divided into 5 sections in 4 main layers.

1. Linux kernel (layer 1)
2. native libraries (middleware) (layer 2)
3. Android Runtime (layer 2)
4. Application Framework (layer 3)
5. Applications (layer 4)

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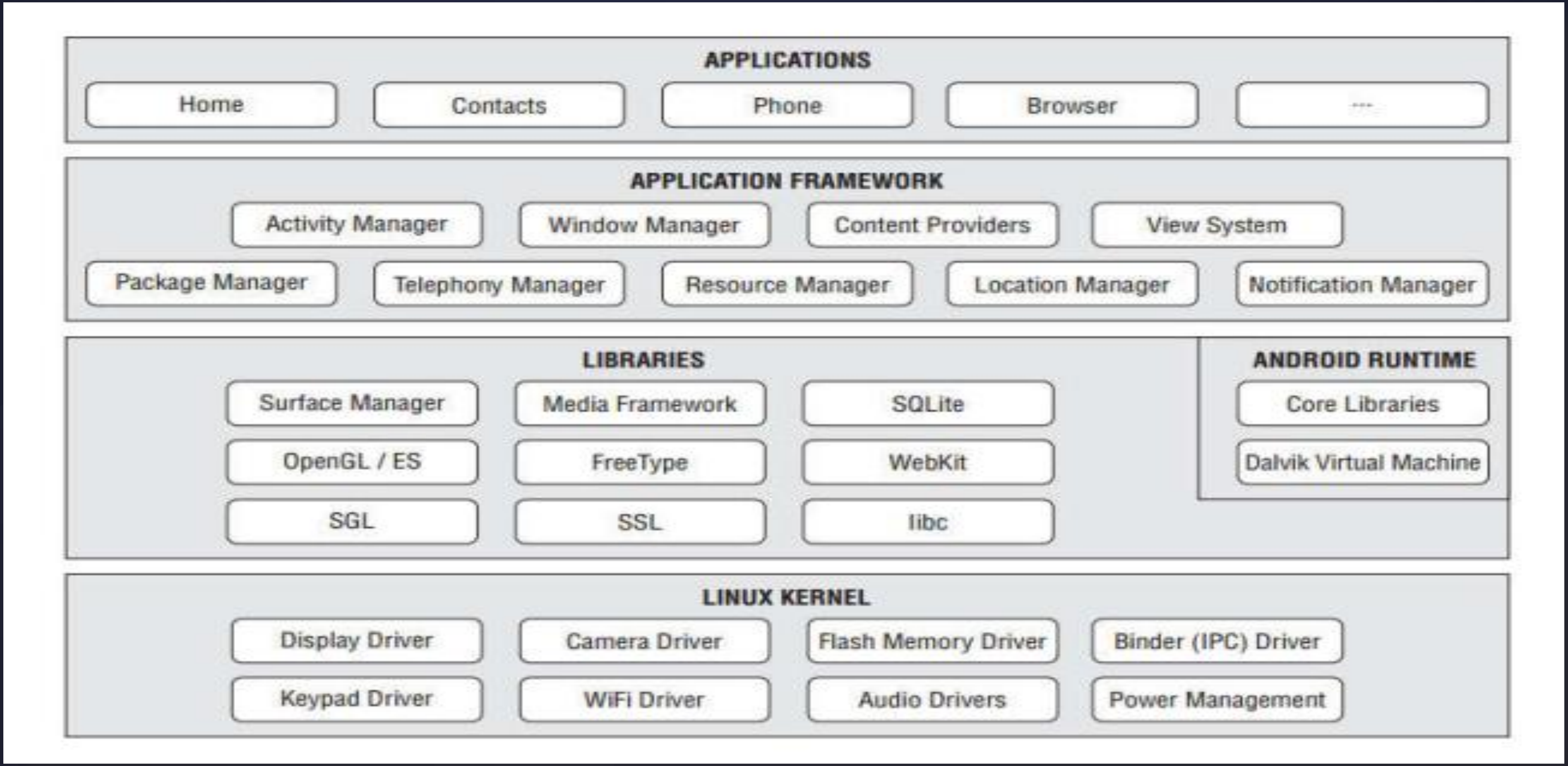
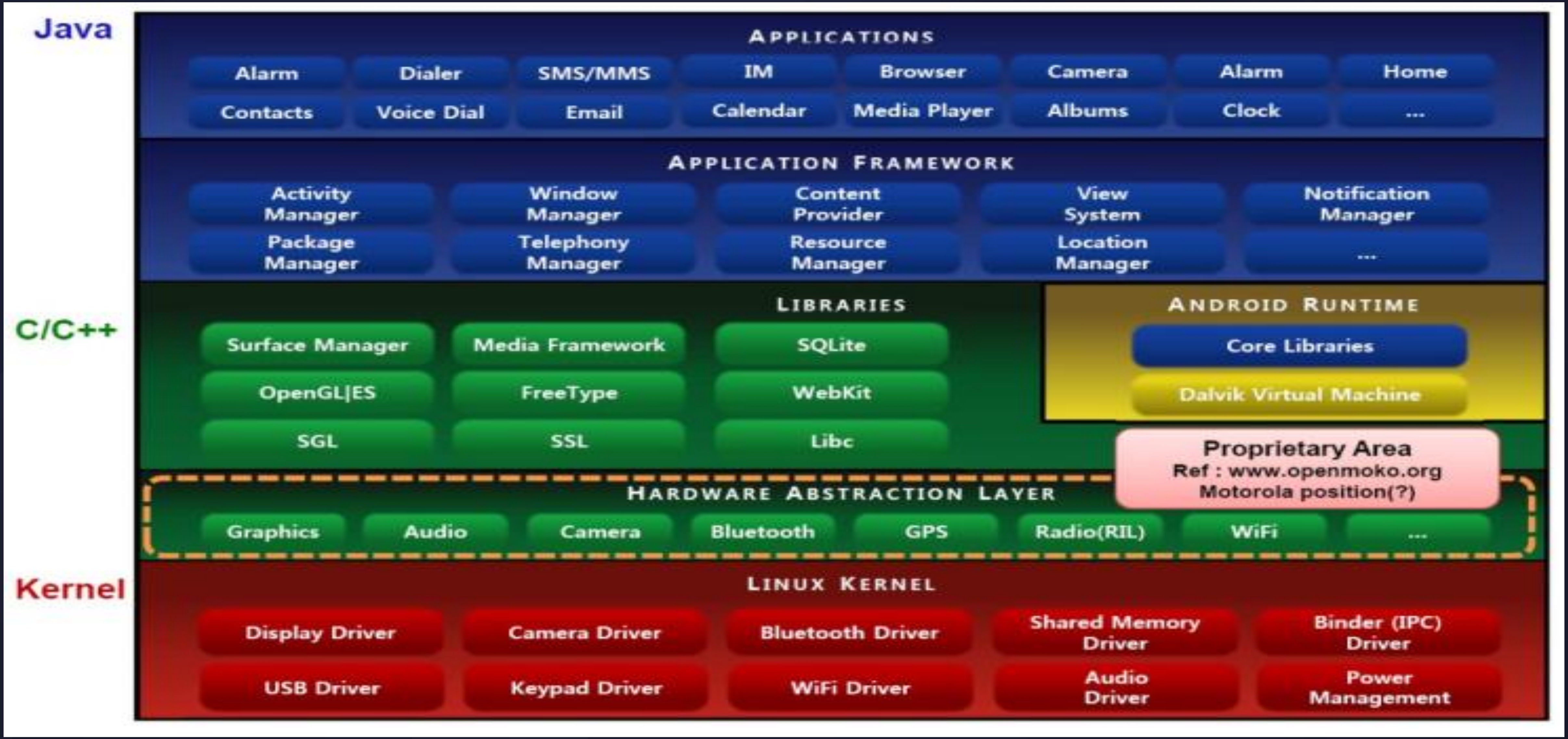

APPLICATIONS
Home
Contacts
Phone
Browser
...
APPLICATION FRAMEWORK
Activity Manager
Window Manager
Content Providers
View System
Package Manager
Telephony Manager
Resource Manager
Location Manager
Notification Manager
LIBRARIES
Surface Manager
Media Framework
SQLite
OpenGL / ES
FreeType
WebKit
SGL
SSL
libc
ANDROID RUNTIME
Core Libraries
Dalvik Virtual Machine
LINUX KERNEL
Display Driver
Camera Driver
Flash Memory Driver
Binder (IPC) Driver
Keypad Driver
WiFi Driver
Audio Drivers
Power Management

OR ..CONT


Java
C/C++
Kernel
APPLICATIONS
Alarm
Dialer
SMS/MMS
IM
Browser
Camera
Alarm
Home
Contacts
Voice Dial
Email
Calendar
Media Player
Albums
Clock
...
APPLICATION FRAMEWORK
Activity Manager
Window Manager
Content Provider
View System
Notification Manager
Package Manager
Telephony Manager
Resource Manager
Location Manager
...
LIBRARIES
Surface Manager
Media Framework
SQLite
OpenGL|ES
FreeType
WebKit
SGL
SSL
Libc
ANDROID RUNTIME
Core Libraries
Dalvik Virtual Machine
Proprietary Area
Ref : www.openmoko.org
Motorola position(?)
HARDWARE ABSTRACTION LAYER
Graphics
Audio
Camera
Bluetooth
GPS
Radio(RIL)
WiFi
...
LINUX KERNEL
Display Driver
Camera Driver
Bluetooth Driver
Shared Memory Driver
Binder (IPC) Driver
USB Driver
Keypad Driver
WiFi Driver
Audio Driver
Power Management

⊙ Linux kernel (layer 1)

- *It is the heart of android architecture that exists at the root of android architecture.*
- *It contains all the low-level device drivers for the various hardware components of an Android device.*
- *It is responsible for*
  - *device drivers and management ,*
  - *power management,*
  - *memory management,*
  - *and resource access.*
- *Users never see Linux sub system*
- *The **adb** shell command opens Linux shell*

Why Linux ? B/c Its

- ***Open source***
- ***Portability*** *(easy to integrate with different hardware)*

  Like Samsung , T-mobile , Sony ,….are all using android platforms since it is easy to port in to different chipsets
- ***Security*** *(Linux is highly secured system)*

  if we Know the Architecture we feel safe to build on something that what we already Know
- ***Essential Features*** *(Linux provides)*
  - *Hardware abstraction layer*
  - *Memory management*
  - *Process management*
  - *Networking*

Native Libraries

On the top of Linux kernel and these contain all the code that provides the main features of an Android OS such as

- ***android.view*** : the fundamental building blocks of application user interfaces
- ***android.opengl*** : a Java interface to the OpenGL ES 3D graphics rendering API
- ***android.database*** : to access data published by content providers and includes SQLite database management classes.
- ***android.widget*** : a rich collection of pre-built user interface components
- ***android.os*** : provides applications with access to standard operating system services including messages, system services and inter-process communication
- ***android.webkit*** :to allow web-browsing capabilities to be built into applications
- ***Media codecs*** offer support for major audio/video codecs
- ***Bionic***, a super fast and small license-friendly libc library optimized for Android
- …..etc

Android Runtime (layer 2)

- There are **core libraries** and **DVM (Dalvik Virtual Machine)** which is responsible to run android application. **DVM** is like JVM but it is optimized for mobile devices. It consumes
  - **less memory,**
  - **battery life and**
  - **provides fast performance**
- Key Dalvik differences:
  - Register-based versus stack-based VM
  - Dalvik runs .dex files
  - More efficient and compact implementation
  - Open source unlike other JVM

○ Application Framework (layer 3)

- *It provides many higher-level services to applications in the form of Java classes. Application developers are allowed to make use of these services in their applications.*
- *It includes **Android API's** such as UI telephony, resources, locations, Content Providers (data) and package managers.*
- *It provides a lot of classes and interfaces for android application development.*
- *Exposes the various capabilities of the Android OS to application developers so that they can make use of them in their applications.*

Applications (layer 4)

- Once developed, Android applications can be packaged easily and sold out either through a store such as **Google Play, SlideME, Opera Mobile Store, Mobango, F-droid** and **the Amazon Appstore.**
- All applications such as messages, contact, settings, games, browsers are using android framework that uses android runtime and libraries.
- Android runtime and native libraries are using Linux kernel.
- as well as applications that you download and install from the Android Market
- There are four main components that can be used within an Android application -
  - **Activities** :They dictate the UI and handle the user interaction to the smart phone screen.
  - **Services** :They handle background processing associated with an application.
  - **Broadcast Receivers:** They handle communication between Android OS and applications.
  - **Content Providers:** They handle data and database management issues

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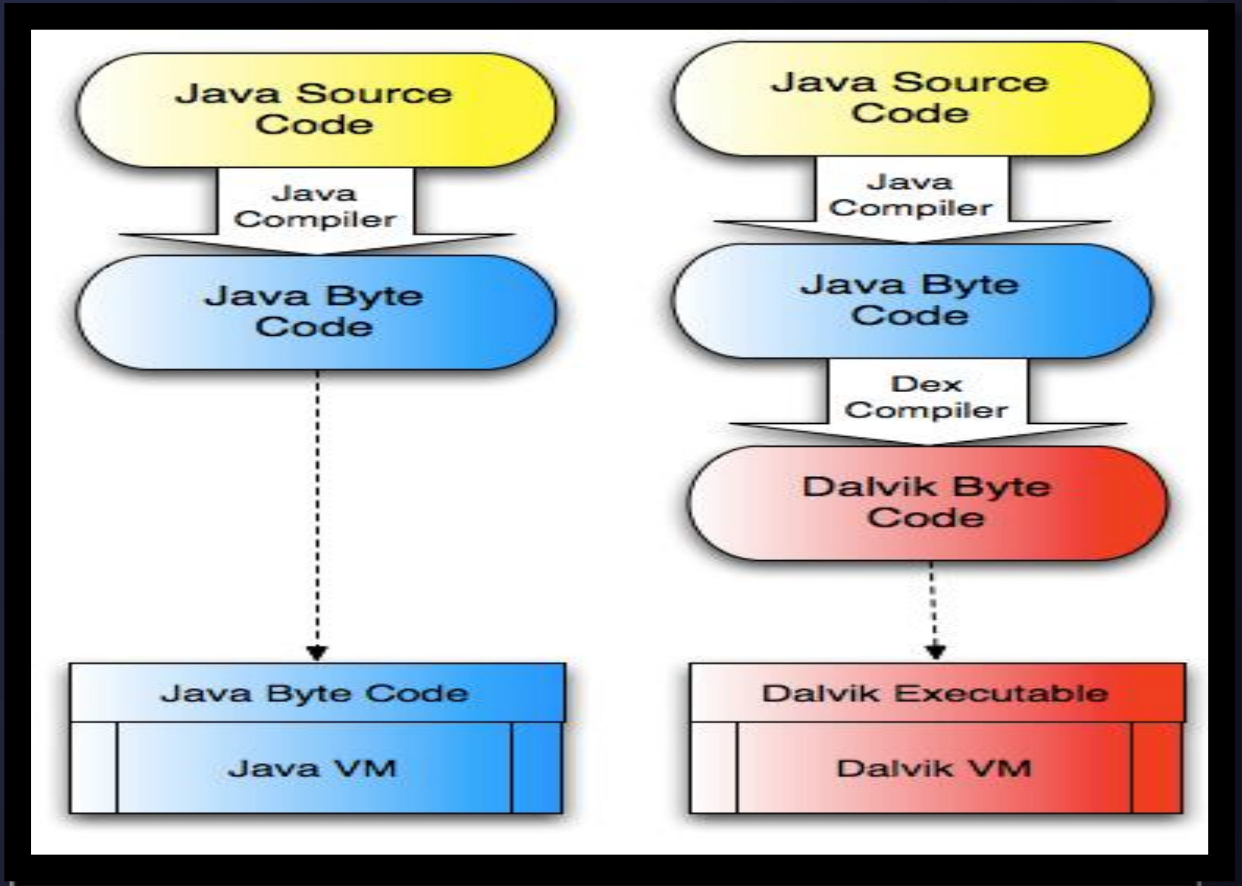
- *Android applications are written in the **java** programming language*
- *The **Android SDK tools compile** the code—along with any data and resource files—into an Android package, an archive file with an **.apk suffix***
- *All the code in a **single .apk file** is considered to be one application*
- *Sign the APK: Signing your APK file ensures its integrity and authenticity. **jarsigner tool** to sign your APK with a digital signature.*
- *Align the APK: Finally, optimize the APK file by aligning its contents on 4-byte boundaries using the **zipalign tool***



java and Android

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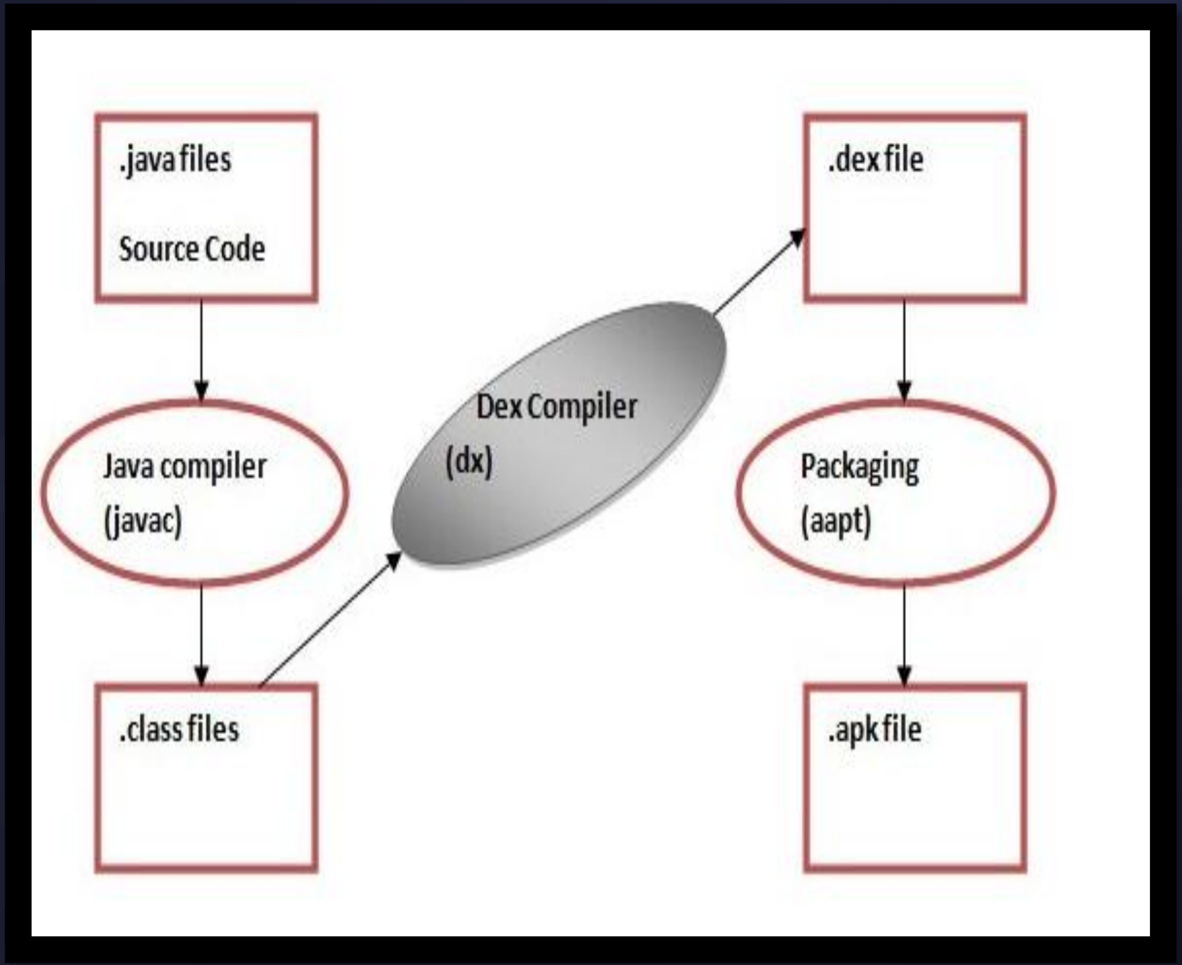
Dalvik Executable(Dex) + Resources = APK



Android and java

- *Android Java*
- *java have different Editions (Java Packaging)*
  - *java SE(Standard Edition)*
  - *Java EE(Enterprise Edition)*
  - *Java ME (Mobile Edition)*
- *but android is not part of these editions*
  - *So why Java*

    :: market issue because tons of java coders in the world

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Android Development Environment

Android Development Environment

Required Tools

- Java JDK5 or JDK6 with JRE
- Android SDK
- Eclipse IDE for Java Developers (optional)
- Android Development Tools (ADT) Eclipse Plugin (optional)

IDE for Android

Android Development Environment

- ***Eclipse***
  - *In the case of Android, the recommended IDE is Eclipse, a multi-language software development environment featuring an extensible plug-in system.*
  - *It can be used to develop various types of applications, using languages such as Java, Ada, C, C++, COBOL, Python, etc.*
- ***Android SDK***
  - *The Android SDK contains a debugger, libraries, an emulator, documentation, sample code, and tutorials.*

- ***Android Development Tools (ADT)***
  - *The Android Development Tools (ADT) plug-in for Eclipse is an extension to the Eclipse IDE that supports the creation and debugging of Android applications.*
- ***Using the ADT***
  - *Create new Android application projects.*
  - *The tools for accessing your Android emulators and devices*
  - *Compile and debug Android applications.*
  - *Export Android applications into Android Packages (APK).*
  - *Create digital certificates for code-signing your APK.*

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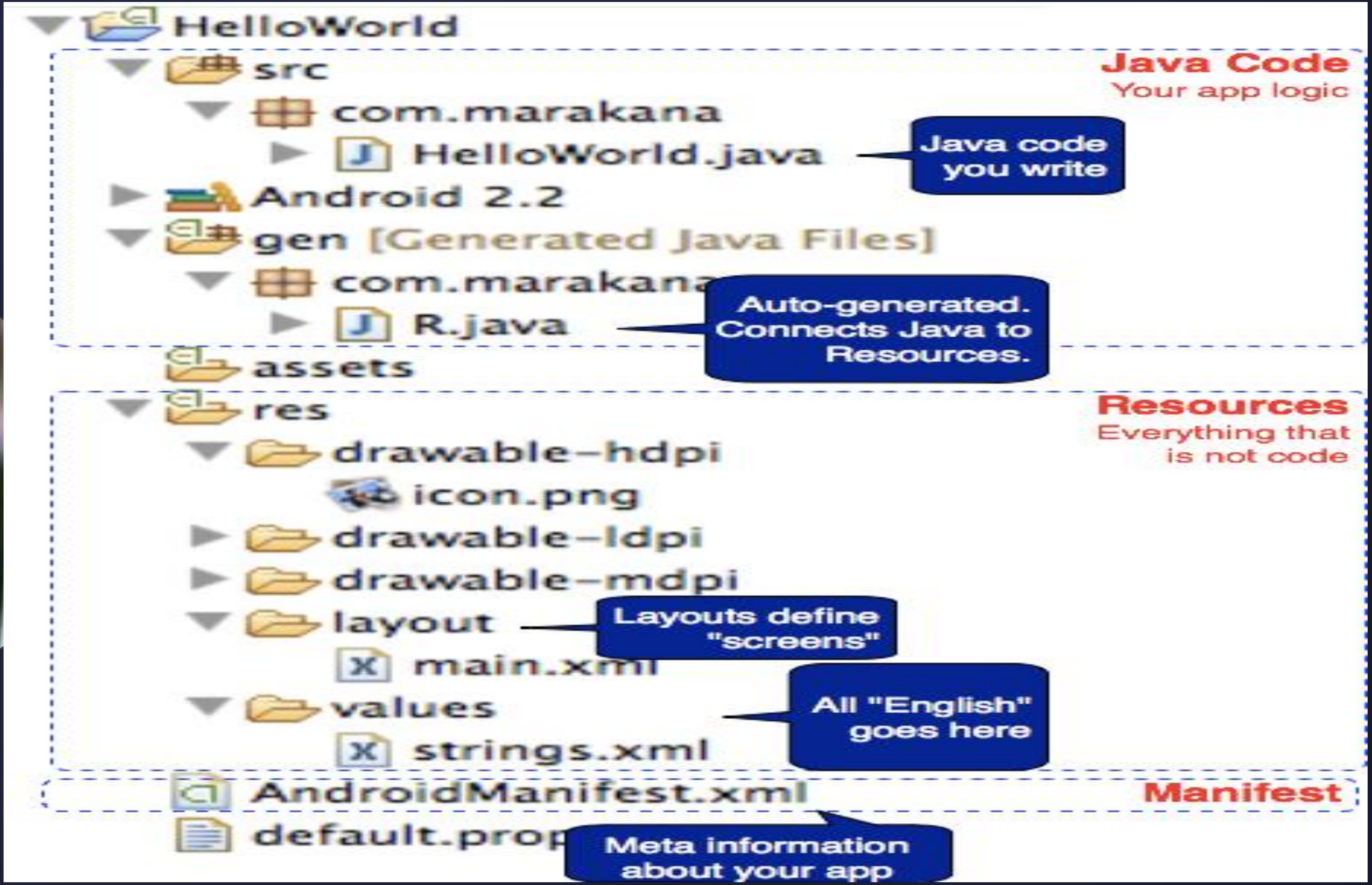
Anatomy of an Android Application

Anatomy of an Android Application



Anatomy of an Android Application

The various folders and their files are as follows:

- **src** — Contains the ***.java** source files for your project. In this example, there is one file, MainActivity.java.
- **Android 4.4 library** — This item contains one file, android.jar, which contains all the class libraries needed for an Android application.
- **gen** — Contains the **R.java** file, a compiler-generated file that references all the resources found in your project. You should not modify this file.
- **assets** — This folder contains all the assets used by your application, such as HTML, text files, databases,
- **res** — This folder contains all the resources used in your application. It also contains a few other subfolders: drawable-<resolution>, layout, and values.
- **AndroidManifest.xml** — This is the manifest file for your Android application. Here you specify the permissions needed by your application, as well as other features (such as intent-filters, receivers, etc.).

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The End !

THANKYOU